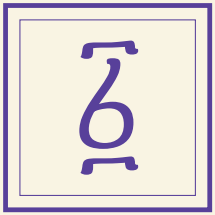
AI in Industry

Introduction and administration

Carlos Cotrini and the teaching team
D-INFK, ETH Zürich
Autumn semester 2026



Section 1

About us

About Carlos



Dr. Carlos Cotrini

- BSc Mathematics, 2009
- MSc Information Security, ETH, 2014
- PhD Information Security, ETH, 2019
- PostDoc in AI and ML, ETH, 2019 to 2021
- Senior scientist, ETH, 2021 to now

Courses at ETH and UZH

- Advanced machine learning, 2019 to now
- Statistical learning theory, 2020 to 2022
- AI in medicine, 2020 to 2022
- Data analysis in physics, 2020 to 2025
- Stochastics and machine learning, 2024 to now
- Building AI/ML applications, 2024 to now
- AI and IT in industry, 2025 to now
- Software engineering fundamentals, 2025 to now
- AI project, 2026 to now
- Applied introduction to machine learning, from 2027

The teaching team



Ivan Angjelovski



Amine Benjelloun Touimi



Ghali Berbich



Francesco Caperna



Konstantinos Chasiotis



Suei-Wen Chen



Luca Chimienti



Monica De Alvaro Mena



Mattia de Martino

The teaching team



Vassili
De Palma



Hassan
El Bouz



Alper
Karadeniz



Joel
Kaufmann



Matina
Lampaki



Angelo
Marano



Samy
Messouak



Elisej
Orsini-Rosenberg



Frederick
Puls

The teaching team



Kangsheng Qi



Lamberto Ragnolini



Winston Rohmann



Daniele Russica



Sanziana Stelea



Christina Tsakanika



Celal Turkmen



Jinlei Zhou

We study how we teach this



Transforming Confusion into Diffusion SIGCSE TS 2026

CER Best Paper (Global) award

- **Full stack ML**: you build the model from the ground up before you reach for a library.
- Trial with $N = 208$: that beat the library first approach by about 10 percent on conceptual understanding ($p = 0.006$), and left participants more curious.



PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning ICER 2026

- *"...multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures."*
- The order we teach things in is itself designed. Trial with $N = 209$: the designed order scored higher ($p = 0.011$, $d = 0.32$) at lower cognitive load.

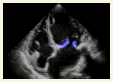
Experimental-study team Dr. Sverrir Thorgeirsson · Stefan Cadar · Sofi Mousia · Umut Demirbas Johannes Schulte-Vels · Nikolaj Murasov · Jonathan Maillefaud · Oscar Jauffret

Some success stories

Oliver Krone, 2025, head of grid and hydro engineering at BKW.

- Entered the Kaggle competition of Advanced machine learning, where students segment the mitral valve of human hearts from echocardiogram videos. Placed 12th out of 100.





The FDD cohort, 2026, a whole class of working professionals.

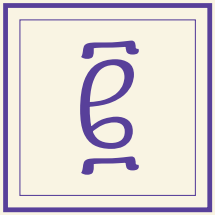
- Handed the first graded project of *Advanced Machine Learning*, an ETH Master's course, with no scaffolding.
- Their 18 teams beat its 136 Master's teams on mean, median and consistency. Top score: an AML grade of 5.95.

Bernhard Obrist, 2025, senior project manager at Aargauische Kantonalbank.

- Built an agent that learns on its own to drive a two dimensional road. Carlos's own agent crashes after 15 seconds.
- Demo: `https://polybox.ethz.ch/index.php/s/raoa3SBoG7cPnT2`







Section 2

Artificial intelligence, what is it?

Four paradigms in science

	Ontological *as it should be*	**Epistemic** *as it is*
Thinking *with brains*	Probability 	Statistics 
Computing *with computers*	Computer science 	**Machine learning** 

Artificial intelligence, a working definition



The design and implementation of systems that perform tasks normally requiring **human intelligence**.

Main approaches

- **Symbolic reasoning:** rules and logic
- **Machine learning:** estimate dependencies from data
- **Deep learning:** machine learning with deep neural networks

Artificial intelligence, a timeline

Foundations



Turing test, 1950



Dartmouth conference, 1956



Rosenblatt's perceptron, 1958

Expert systems, AI winter

LISP, 1958

DENDRAL and MYCIN, 1970s

Statistical machine learning



Vapnik's statistical learning, 1982



IBM Watson, 2011

Deep learning



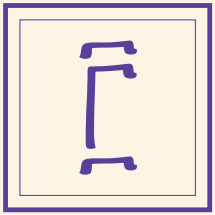
AlexNet, 2012



AlphaGo, 2016



ChatGPT, 2022



Section 3

Scope and learning goals

What you learn in this course

Perspectives of this course

Phenomenological: How AI technologies behave.

Constructive: How to compose new systems out of AI technologies.

+

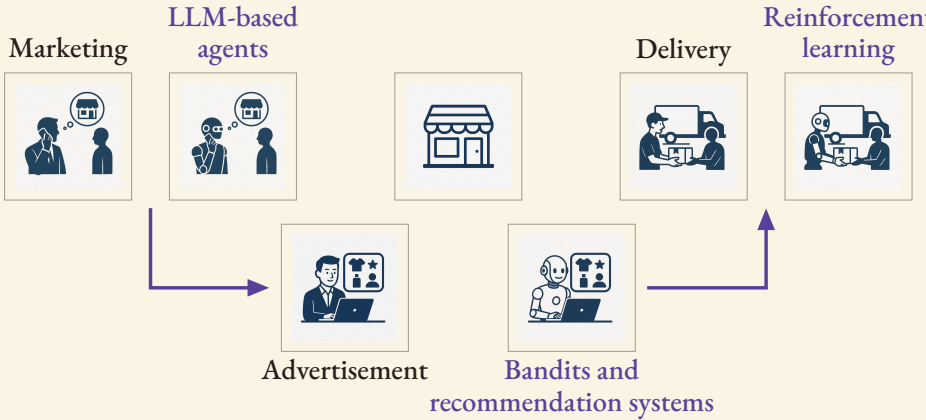
Perspectives of the other modules

Theoretical: How AI works and how you can create new AI technologies.

Practical: How to use AI technologies effectively.

Foundational: What are the principles and methods of AI.

The project: AI-powered retail transformation



The project: AI-powered retail transformation



You will apply the concepts of this course to transform a retail store into an AI-powered business.

- **LLMs** for marketing, using prompt optimization techniques.
- **Bandits and recommendation systems** for targeted advertisement and customer profiling.
- **Reinforcement learning** for optimizing delivery strategies.

Part 1: LLM-based marketing agent

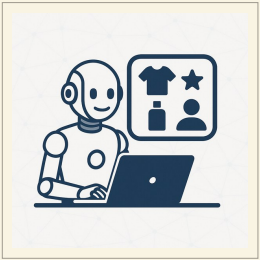




You learn how to use automatic prompt optimization methods to create an LLM-based marketing agent that iteratively learns to improve its communication style to attract more customers.

Part 2: Recommendation systems

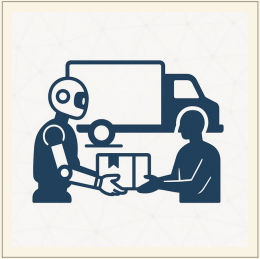




You learn to implement matrix-factorization methods for creating recommendation systems and targeted advertisement.

Part 3: Reinforcement learning for delivery optimization





You learn to use reinforcement learning to optimize the delivery of orders to customers.

Learning objectives, Part 1: LLMs

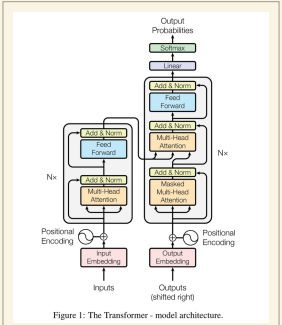
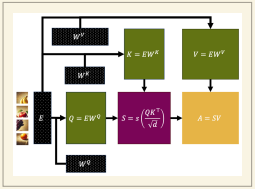
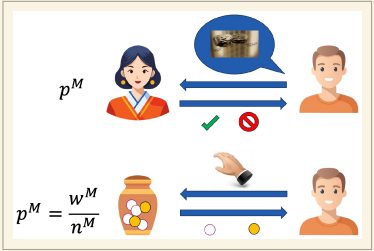


Figure 1: The Transformer - model architecture.



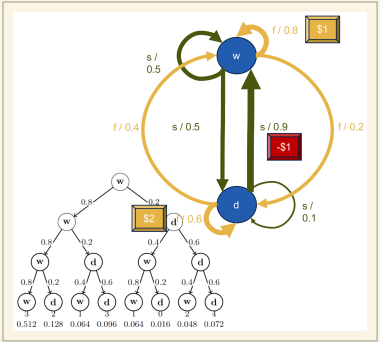
- To understand how LLMs work.
- To be able to identify limits in reasoning in LLMs.
- To design systems that enactively use LLMs to solve complex tasks.
- **Project task:** To design data-driven bootstrapped marketing strategies through LLMs.

Learning objectives, Part 2: Bandits and recommendations

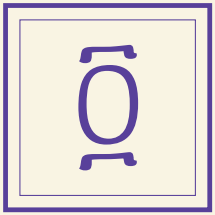


- To understand how to develop algorithms for marketing and targeted advertisement.
- To understand how recommendation systems work as matrix factorization problems.
- To design and implement algorithms for targeted advertisement and recommendation systems.
- **Project task:** To develop strategies for comparing and replacing ML-based strategies based on performance.

Learning objectives, Part 3: Reinforcement learning



- To model reward-based systems via agents, environments, and Markov decision processes.
- To familiarize with basic optimization routines like Robbins-Monro and Q-learning.
- To develop reinforcement-learning agents that effectively maximize rewards in small-scale environments.
- **Project task:** To develop and implement a routing algorithm for an autonomous delivery truck based on reinforcement-learning.



Section 4

Roadmap

Roadmap

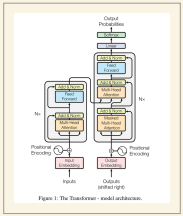


Figure 1: The Transformer - model architecture.

Transformer

Prediction

Compression

Intelligence

Language models

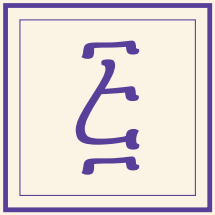
Large language models

Agents

Test-time computation

- Mesa-optimization
- Prompt engineering
- **OPRO**: Optimization via prompting

Awareness and self-preservation



Section 5

Administration

Please read these slides carefully.

The course website

https://eth-ainit-hs26.github.io/AINIT-PAGE/

- The agenda of each weekend, Friday and Saturday, with its slides and exercises.
- The course calendar and the booking page for office hours.
- The lecture recordings. The password is in the course Moodle.



Scan it to open the site on your phone

Passing the course

To pass the course you must do all of the following.

- Complete the **three parts** of the project and submit your code on Moodle:
  `https://moodle-app2.let.ethz.ch/course/view.php?id=28801`
- Defend your project to one of the teaching assistants in a one hour presentation at the end of the course.
- Pass the **three quizzes** on Moodle, which ask about each of the weekends. You may retake a quiz as often as you need, but you must reach at least 80 percent of the points.

Everything above must be finished before the grand deadline

Sunday 31 January 2027, 23:59

The project

- The project has three parts, one per weekend: an **LLM-based marketing agent**, a **recommender system**, and an **RL-based delivery truck**.
- We release each part in order, after each weekend. We recommend that you finish each part before the next weekend.
- Projects are solved **in teams of three** wherever possible.
- Form a team, then choose the teaching assistant you will present your final implementation to, and mark your choice on Moodle. Arrange a 60 minute Zoom meeting with that assistant and defend your implementation before the grand deadline. **Do this early**, so that you both find a suitable time.

Weekend	Dates in 2026	Project part
W1, Large language models	16 and 17 October	LLM-based marketing agent
W2, Representation learning	13 and 14 November	Recommender system
W3, Reinforcement learning	11 and 12 December	RL-based delivery truck

Google Colab

- We recommend Google Colab for this course:
  `https://colab.research.google.com/`
- You get a Google Colab Pro license, valid until the grand deadline.
- To have it, you need to set up a Google Workspace. The steps are in section 5 of the technical onboarding:
  Technical onboarding, setting up Colab



Scan it to open the onboarding

Office hours

- The teaching team holds office hours over Zoom between the weekends.
- **Tuesday, Wednesday and Thursday, 18:00 to 20:00**, in 30 minute slots.
- **Book a slot on the course website.** Enter your name, then click an open slot. Anyone can cancel a booking.
- Two teaching assistants are on duty each day. The meeting link is the one that assistant posts, and it appears under their name on the same page.



Scan it to book a slot

`https://eth-ainit-hs26.github.io/AINIT-PAGE/#/office-hours`



Questions?

eth-ainit-hs26.github.io/AINIT-PAGE